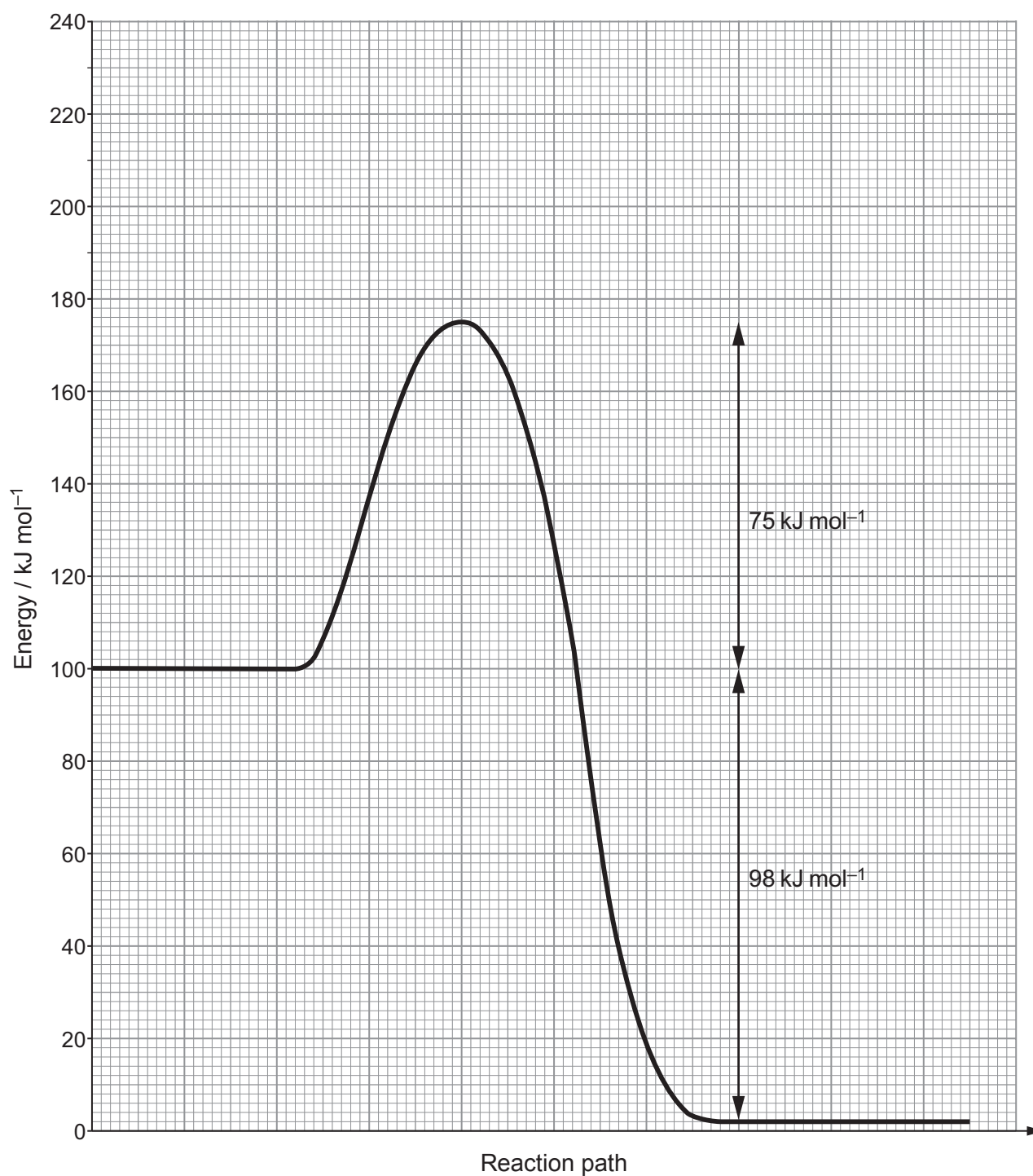


11. Hydrogen peroxide, H_2O_2 , decomposes slowly at 300 K to form water and oxygen gas.



The reaction profile for this reaction is shown below.



- (a) The rate of decomposition of hydrogen peroxide can be influenced by a range of catalysts. The activation energy when using MnO_2 as a catalyst is 58 kJ mol^{-1} .

Draw the reaction profile for the catalysed reaction **on the grid opposite**. [1]

- (b) Another catalyst was used and this gave a value for the rate constant, k , of $1.68 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ and the frequency factor, A , of $1.41 \times 10^4 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ at a temperature of 300 K.

- (i) State the Arrhenius equation. [1]

- (ii) Calculate the activation energy using this catalyst and hence state whether it is a more effective catalyst than MnO_2 . [3]

Activation energy = kJ mol^{-1}

- (c) The standard enthalpy change of formation, $\Delta_f H^\theta$, of water is -286 kJ mol^{-1} .

Use this information and the graph to calculate the standard enthalpy change of formation of hydrogen peroxide. [2]

$\Delta_f H^\theta(\text{H}_2\text{O}_2) = \text{.....} \text{ kJ mol}^{-1}$

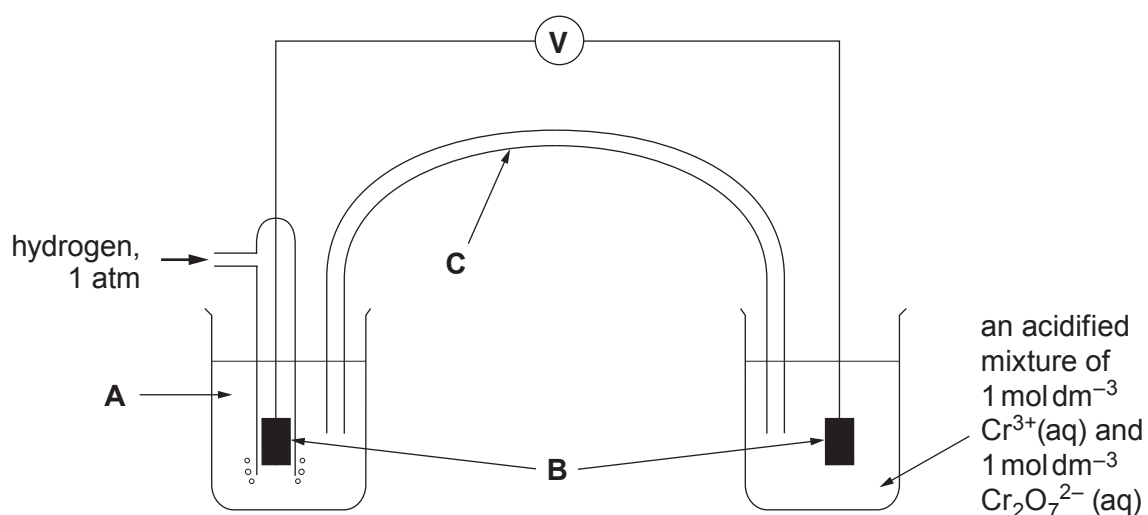
- (d) State whether you would expect the entropy change for the decomposition of hydrogen peroxide to be positive or negative. Give a reason for your answer. [1]



(e) One way of assessing whether a reaction is feasible is to use standard electrode potentials.

	Standard electrode potential / V
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	+1.77
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1.33

- (i) The apparatus below can be used to measure the standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$ half-cell.



- I. State what is represented by **A** and **B** on the diagram. [1]

A

B

- II. **On the diagram** show the direction of flow of electrons in the external circuit. [1]



III. State what is represented by **C** on the diagram and state its function. [1]

.....

.....

IV. The concentrations of both $\text{Cr}_2\text{O}_7^{2-}$ and Cr^{3+} ions are 1 mol dm^{-3} . State and explain how the value shown on the high resistance voltmeter would change if the concentration of the Cr^{3+} ions were increased whilst the concentration of the $\text{Cr}_2\text{O}_7^{2-}$ was left unchanged. [2]

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(ii) It is suggested that hydrogen peroxide could be used to oxidise Cr^{3+} ions in acidic solution to form dichromate ions.

I. Write an equation for this proposed reaction. [1]

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II. Use the standard electrode potential values given to predict whether this reaction is feasible. [2]

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III. Another method of finding whether a reaction is feasible is to use the Gibbs free energy calculated from standard enthalpy of formation and standard entropy values.

State, giving a reason, whether Gibbs free energies or electrochemical methods are more appropriate for finding whether this reaction is feasible. [2]

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